

Chapter 6

Conclusions and Outlook

This thesis investigated whether the volatility of an organic polymer-electrolyte memristive device can support temporal computation. ?????? establish the device response, diagnose its reproducibility, and measure how composition and chemistry change its dynamics. ?? evaluates behavioural models of those dynamics on temporal-computing tasks. This chapter summarises the results, their limitations, and the experiments needed to test them in hardware.

6.1 Summary of Contributions

The contributions fall into four parts: a proof-of-concept device, a reproducibility and fabrication-record study, a comparative composition and chemistry study, and temporal-computing simulations. Their evidentiary strength differs because only the composition grid is replicated across devices.

A characterised synaptic platform (??). A reversible two-terminal organic memristive platform combines *Super Yellow*, *Hybrane*, and LiOTf

in a solution-processed active layer between ITO and Ag electrodes. Across assay-specific junctions sharing that formulation and architecture, it supports analogue I–V hysteresis (an 85 % conductance increase from the first to the second sweep at 1 V, growing ~ 15 -fold over ten cycles), reversible potentiation and depression exceeding 200 % at an energy cost of order 50 nJ per event, and two distinguishable retention regimes. Standard-exponential fits reported in the published SI give $\tau_S \approx 2.5\text{ s}$ – 3 s for short-term memory and $\tau_L \approx 4.7\text{ s}$ for long-term memory. The formulation also supports a four-state excitatory post-synaptic current read-out and an asymmetric Hebbian STDP window with a time constant of 85 ms–90 ms. Mobility differences between weakly coordinated triflate and strongly coordinated lithium in the oxygen-rich host provide a hard–soft acid–base interpretation of the two retention regimes. The assays were independently peer reviewed [1], but they do not constitute a population study; ?? gives the specimen mapping.

A reproducibility diagnosis and a fabrication-record methodology (??). Successive batches of nominally identical *Hybrane* devices changed from low-current, high-area, multilevel behaviour to a high-conductivity response without a usable switching window. Per-device weighting, restriction to a standard protocol, sweep-amplitude matching, and recovery-tail sensitivity confirmed a multi-feature change: the on–off ratio fell from 2.44 to 1.21, normalised hysteresis area from 0.235 to 0.027, and potentiation at 3 V changed from +17.9 % to –2.1 %. Conduction also became more ohmic at a matched 1.2 V. The change followed fabrication date rather than device age, because finished *Hybrane* devices retained their characteristics over weeks. The evidence points to ageing of the *Hybrane* stock, with *Super Yellow* excluded, under a moisture-driven hydrolysis hypothesis supported by an annealing-recovery test. Replacing *Hybrane* with poly(ethylene oxide) increased the normalised window area from 0.26 to 0.42 at 3 V and maintained usable switching across roughly

two and a half years. The study also produced the fabrication-record standard, normalised characterisation procedure, automated feature pipeline, and trend-analysis tools used in the later chapters.

Compositional and chemical control of the dynamics (??). The replicated *Super Yellow*/PEO/LiOTf/Ag grid shows that composition tunes fading-memory time over roughly 2 s–20 s; higher ion-transport-polymer content gives a weaker, faster-relaxing response. Host (PEO versus TMPE) and anion (triflate versus TFSI) extend the range in the illustrative chemistry survey. Cation identity also shifts the dynamics, but the triflate series runs opposite to the hypothesised $\text{Li}^+ > \text{Na}^+ > \text{K}^+$ order and changes again with the anion. The available data therefore do not establish a transferable cation law. Drive amplitude changes the apparent timescale, so quantitative comparisons require a matched protocol. Equilibrium impedance falls along the composition axis and tracks fading-memory time. ATR, XRD, and UV-Vis associate the host effect with short-range ether order in an X-ray-amorphous, salt-dissolved film without a shift in the *Super Yellow* optical gap. Because the PEO fraction changes both fading memory and input nonlinearity, each composition defines a paired operating point for the simulations of ??.

A temporal-computing demonstration from measured dynamics (??). Compact behavioural models ($\phi \otimes \lambda \otimes f$) fitted to the ?? measurements were evaluated with linear read-outs. A heterogeneous composition bank raises total linear memory capacity from 4.2 to 6.4 (a factor of 1.54 ± 0.10 , positive in all ten seed pairs) and total information-processing capacity from 7.1 to 11.6. Seeds quantify algorithmic sensitivity rather than experimental replication. The nonzero nonlinear capacity of the homogeneous bank shows that the compressive write contributes a usable nonlinear response. The heterogeneous bank improves multi-lag

reconstruction of physiological streams. On WESAD, the models reach leave-one-subject-out macro-F1 scores of 0.894 for binary stress and 0.758 for three-class streaming classification. On CASE, a multi-timescale filter improves continuous valence/arousal tracking by +0.176 concordance correlation over a single timescale (27/30 subjects). Timescale diversity does not separately improve final-label classification when one slow timescale dominates. These results are simulations of measured device models, not measurements from a wired reservoir.

6.2 Answers to the Thesis Questions

The results answer the three questions posed in ??.

First: realising and understanding an exemplar. *Can a reversible two-terminal organic memristive platform made from a solution-processed polymer-electrolyte composite support analogue switching, potentiation and depression, distinct retention regimes, EPSC, and STDP responses, and can cation-anion mobility asymmetry account for the retention regimes?* The common formulation and architecture of ??, fabricated by solution processing and thermal evaporation, support these responses across assay-specific junctions. Differential displacement and relaxation of triflate and lithium provide a mechanism consistent with the two retention regimes. Because the extended assays are not a replicated device-population comparison, the mechanism remains a hypothesis rather than a population-level law.

Second: composition as the quantitative control, with chemistry as a sample-limited survey. *How do the dynamics depend on composition, and does moving the cation along the Li^+ , Na^+ , K^+ series shift the timescale as coordination chemistry suggests?* The replicated PEO/LiOTf

grid shows a monotonic composition dependence: the ratio tunes fading-memory time over roughly an order of magnitude, with switching window and potentiation strength moving in the same direction. Cation identity shifts the dynamics, but the triflate retention series runs opposite to the predicted $\text{Li}^+ > \text{Na}^+ > \text{K}^+$ order and changes again with the anion. Sample size and protocol/electrode confounds prevent a transferable cation law. Coordination chemistry remains a qualitative interpretation rather than a quantitative result.

Third: utility for temporal computing. *Can the volatile response be used for temporal computation, can composition supply task-matched timescales, and does measured variability broaden the response of a heterogeneous bank?* In the simulations of ??, composition supplies fading-memory times that overlap peripheral affective-signal bands. A heterogeneous bank broadens memory and information-processing capacity, and device-to-device variation contributes additional state diversity. The benefit depends on the task: the lead composition is sufficient for a single-timescale target, and the heterogeneous increment is not resolved for the sustained WESAD labels.

6.3 Benchmarking Against the Right Reference Class

Because the device state relaxes over seconds, temporal and event-driven neuromorphic systems provide the relevant reference class [2–4]. Non-volatile DRAM, SRAM, and crossbar ReRAM serve different storage functions. Current physical-reservoir comparisons must also separate the physical node, input encoding, output layer, and network architecture [5]; task scores from different layers and protocols are not a common

leaderboard.

Against physical reservoirs. Physical reservoir computing has been demonstrated in dynamic memristors [6, 7], organic electrochemical devices [8], and delay-coupled single nodes [9]. More recent systems add direct electrical tuning in two-terminal oxide devices [10], gate- and wavelength-controlled organic transistor dynamics [11], and an organic electrochemical reservoir with a physical device read-out [12]. These systems require fading memory and nonlinearity [4, 13]. The present devices have measured fading memory, while the fitted models have nonzero nonlinear information-processing capacity. Their sub-kilohertz response overlaps many physiological and environmental signals without time rescaling. Composition changes the fading-memory time over 2 s–20 s, and the corresponding heterogeneous model bank broadens memory capacity by a factor of 1.54.

Against organic neuromorphic devices. Much of organic neuromorphic electronics targets linear, symmetric, non-volatile conductance updates for weight storage, as in the ENODE family [14–16]. The devices studied here use ion-relaxation timescales for temporal processing and are compared with ionic and electrolyte-gated organic neuromorphic systems [8, 17].

Placement and evidence boundary. The experimentally supported claim is that composition tunes seconds-scale fading memory in this two-terminal organic platform. The computational claim is that banks of behavioural models fitted to those measurements can support the reported benchmarks and physiological tasks. The application scores are *in silico* and do not establish the performance of a wired array; nor is priority claimed for a first tunable two-terminal reservoir. The simulated reservoir

Table 6.1: Recent directly relevant reference points for tunable physical reservoirs. Temporal ranges are the values reported under each study’s own extraction protocol; tasks, read-outs, and metrics differ, so the table calibrates scope rather than ranking performance. “Physical” means that measured device states formed the reservoir; it does not imply that every input/output operation was analogue or on-chip.

Study	Node and terminals	Tuning handle	Reported temporal range	Reservoir evidence and task scope
This thesis	<i>Super Yelow</i> /PEO/LiOTf organic composite; two-terminal	PEO/salt composition; drive amplitude	2 s–20 s across the replicated composition grid	Single-node dynamics measured; heterogeneous banks and WESAD/CASE tasks simulated; no wired reservoir
Gao et al. (2024) [11]	Organic vertical field-effect transistor; three-terminal optoelectronic	Input wavelength and gate bias	0.005 s–13.2 s with optical input and gate bias	Physical grouped-reservoir states; image recognition and time-series prediction
Han et al. (2025) [12]	Solid-electrolyte OECT; three-terminal	Ethylene-glycol content selects volatile or non-volatile operation	No common fitted- τ range reported	Physical reservoir and physical device read-out; three-class human-activity classification
Lee et al. (2025) [10]	Pt/HfO ₂ /TiN memristor; two-terminal	Inter-pulse base voltage	Fitted τ : 0.92 ms to 123 ms; inputs 0.14 ms–100 ms	Physical reservoir states; N-MNIST classification and Mackey–Glass prediction

contains independent, non-recurrent nodes; its absolute NARMA errors are high and its capacities remain below the node-count bound. Recurrent or delay-feedback connectivity could raise these values. The evidence chain therefore ends at measured dynamics plus model-based task validation; the missing end-to-end test is a fabricated, composition-diverse reservoir under the same tasks.

6.4 Limitations of the Evidence Base

The principal limitations are as follows.

1. **The extended synaptic protocols are proof-of-concept rather than population-level.** EPSC, separated short-term memory/long-term memory retention, and STDP exist only for the ?? *Hybrane* formulation; some specimen identifiers are not resolved in the recovered archive, and the available multi-junction data share one substrate. They function as formulation-level lithium priors and checks in the later chapters and were not extended to comparative status across the PEO family.
2. **Only composition is replicated.** The quantitative core is the *Super Yellow*/PEO/LiOTf/Ag composition grid. The host, anion, and cation comparisons rest on one or two screened devices per matched cell and are illustrative. The electrode contrast is also illustrative and partly confounded with fabrication generation; its negative-polarity measurements are single-composition and uncured, and its frequency-domain measurements rest on one batch with substrate-confounded cations. No powered material law, including a $\text{Li}^+ > \text{Na}^+ > \text{K}^+$ ordering, is claimed.
3. **Protocol-specific timescales.** Because the drive amplitude co-sets the apparent fading-memory time, absolute retention values

are protocol-specific, and the supra-threshold read used in the comparative corpus differs from the sub-threshold read of the proof-of-concept device.

4. **The composition assumption behind the models.** Pulse integration and state relaxation were characterised separately, at different write amplitudes (potentiation at 3 V, decay at 4 V) and one inter-pulse cadence. Combining them in the behavioural models assumes that the two regimes are compatible. A coupling-sensitivity control uses the direct estimate from the gold/TMPE corpus ($\rho = +0.46$, mapping to $\kappa^* = +0.18$); the heterogeneous bank retains its capacity advantage at that value. A matched protocol is still needed for the lead composition.
5. **The application results are in silico and only partly cross-corpus.** Every temporal-computing result is a simulation of measured device models with a linear read-out; no wired reservoir was built. The physiological-context and affective-label results use WE-SAD as the main corpus, and heterogeneity does not separately improve its sustained labels. Noise robustness replicates on the PhysioNet Non-EEG corpus. Continuous valence and arousal are tested on CASE, where the device’s nonlinear encoding does not exceed a linear multi- τ filter with the same timescale spread. Broader multi-corpus validation remains future work.
6. **Material stability remains open.** Devices were characterised unencapsulated on a ~ 300 h ambient shelf scale; cycling endurance, read-disturb under continuous pulsing, and long-run drift under encapsulation were not systematically studied, and device ageing was not modelled in the comparative corpus.

6.5 Outlook and Future Directions

The immediate experiments address the model assumptions and sample-limited comparisons; longer-term work concerns stability and hardware integration.

Matched pulse protocol. A matched-amplitude pulse train should vary pulse count and inter-pulse interval together for one composition, electrode, and read protocol. The gold/TMPE negative-polarity corpus of ?? already records integration and relaxation within single runs and finds interval-dependent potentiation and drive-depth-dependent relaxation. Repeating this measurement on the lead silver composition with a sub-threshold read would supply the missing joint parameters for the behavioural model.

Replicated chemistry survey. A cation series in one host and anion, at fixed protocol and electrode, with a sub-threshold read and at least three replicates per cell, would test the $\text{Li}^+ > \text{Na}^+ > \text{K}^+$ hypothesis at adequate power. The silver TMPE retention follows the opposite order, $\text{K} > \text{Na} > \text{Li}$; the gold steady-state window is flat; and the gold negative-polarity data resolve no order. The lock-in corpus contains one cation-ordered candidate: a $\approx 1.4 \times \text{K} > \text{Na} > \text{Li}$ junction-capacitance trend at one substrate per salt (??). Profilometry of the existing substrates and a replicated series could test whether this trend is chemical or geometric. Chemistry- and bias-resolved impedance and a generation-controlled silver-versus-gold split would strengthen the microscopic and electrode comparisons.

Test the degradation chemistry. The moisture-driven hydrolytic-aging hypothesis for the *Hybrane* stock (??) cannot be tested directly because the aged consumable no longer exists in its degraded state. FTIR,

GPC, Karl-Fischer/TGA, and EIS on retained aged-versus-fresh stock would test the hypothesis in future material campaigns. The fabrication-record methodology now records reagent identity and history.

Engineer the host and salt for finer timescale control. Modifying the polyether host and salt could extend fading memory toward the minutes-scale tonic band and separate memory time from the input nonlinearity that composition currently changes with it.

Stabilise for long-run inference. Encapsulation and systematic measurements of ageing, endurance, and read disturb are required to assess continuous operation and remove uncontrolled ageing from later comparisons.

Build the hardware reservoir. A wired bank of composition-diverse PEO/LiOTf devices with an on-chip linear read-out would test the simulations directly. Integration toward a 1T1M-style array and operation on an affective stream would measure whether the predicted benefit of timescale heterogeneity survives in hardware.

Physiological sensor interfaces. Solution processing, mechanical flexibility, biocompatible chemistry, and seconds-scale dynamics make these devices candidates for skin-conformal sensing and inference. In the frequency-domain measurements of ??, a DC-biased junction acts as a voltage-programmable low-pass amplifier below 10 Hz. A hardware experiment could test whether one junction can combine pre-amplification and temporal filtering at a physiological sensor interface.

6.6 Closing Remarks

This thesis shows that composition tunes the seconds-scale fading memory of polymer-electrolyte memristive composites and that behavioural models fitted to those measurements can process temporal signals. The replicated evidence supports composition-dependent control; host, anion, cation, and electrode effects remain illustrative, and the simulations do not replace a wired reservoir. The next experimental priorities are a matched-amplitude pulse protocol, a replicated chemistry series, and a hardware bank of composition-diverse devices.

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